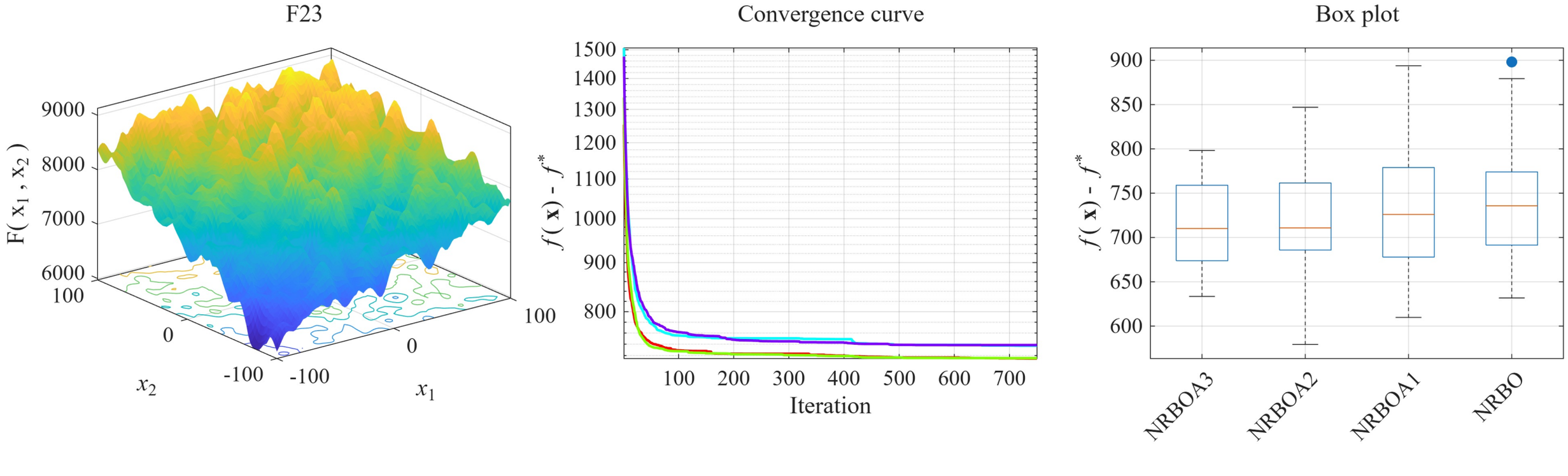

F23
F(x1 , x2)
9000
8000
7000
6000
100
0
-100
x2
-100
0
100
x1
Convergence curve
f(x) - f*
1500
1400
1300
1200
1100
1000
900
800
100
200
300
400
500
600
700
Iteration
Box plot
f(x) - f*
900
850
800
750
700
650
600
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO